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GENOME REPLICATION AND REGULATION OF VIRAL GENE EXPRESSION

Insights into Two-Metal-Ion Catalytic Mechanism of Cap-Snatching Endonuclease of Ebinur Lake Virus in *Bunyavirales*

Wenhua Kuang,[a] Huanyu Zhang,[b] Yan Cai,[c,e] Guilin Zhang,[d] Fei Deng,[b] Hailong Li,[d] Zhihong Hu,[b] Yu Guo,[c] Manli Wang,[b] Yiwu Zhou,[a] Peng Gong[f,g]

[a]Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China
[b]State Key Laboratory of Virology, Wuhan Institute of Virology Chinese Academy of Sciences, Wuhan, Hubei, China
[c]State Key Laboratory of Medicinal Chemical Biology, College of Life Sciences, Nankai University, Tianjin, China
[d]Center for Disease Control and Prevention of Xinjiang Military Command, Urumqi, Xinjiang, China
[e]Tianjin International Joint Academy of Biotechnology and Medicine, Tianjin, China
[f]Key Laboratory of Special Pathogens and Biosafety, Wuhan Institute of Virology, Center for Biosafety Mega-Science, Chinese Academy of Sciences, Wuhan, Hubei, China
[g]Drug Discovery Center for Infectious Diseases, Nankai University, Tianjin, China

ABSTRACT The cap-snatching endonuclease (EN) of segmented negative-strand RNA viruses (sNSVs) produces short capped primers for viral transcription by cleaving the host mRNAs. EN requires divalent metals as cofactors for nucleic acid substrates cleavage; however, the detailed mechanism of metal ion-dependent catalysis of ENs remains obscure. In this work, we reported the EN crystal structure of the Ebinur Lake virus (EBIV), an emerging mosquito-borne orthobunyavirus, and investigated its enzymatic properties and metal ion-based catalytic mechanism. *In vitro* biochemical data showed that EBIV EN is a specific RNA nuclease and prefers to cleave unstructured uridine-rich ssRNA. Structural comparison indicated that the overall structural architecture of EBIV EN is similar to that of other sNSV ENs, while the detailed active site configuration including the binding state of metal ions and the conformation of the LA/LB loop pair is different. Based on sequence conservation analysis, nine active site mutants were constructed, and seven crystal structures of them were determined. Mutations of active site residues associated with the two metal ions (Mn1 and Mn2) coordination abolished EN activity. Crystallographic analyses further revealed that none of these mutants bound two metal ions simultaneously in the active site. Importantly, we found that the perturbation of Mn1-coordination (metal site 1), resulted in the enhancement or elimination of Mn2-coordination (metal site 2). Taken together, our data provide structural evidence to support the two-metal-ion catalytic mechanism of EBIV EN and the correlation of metal binding at the two binding sites, which may be commonly shared by bunyaviruses or other sNSVs.

IMPORTANCE The viral endonucleases (ENs) encoded by bunyaviruses and orthomyxoviruses play an essential role in initiating transcription by “snatching” capped primers from the host mRNAs. These ENs are metal-ion-dependent nucleases; however, the details of their catalytic mechanism remain elusive. Here, we reported high-resolution crystal structures of the wild-type and mutant ENs of a novel bunyavirus, the Ebinur Lake virus (EBIV), and revealed the structure and function relationship of EN. The EBIV EN exhibited differences in the details of active site structure compared to its homologues. Our data provided structural evidence to support a two-metal-ion catalytic mechanism of EBIV EN, and found the correlation of metal binding at both binding sites, which might reflect the dynamic structural properties that correlate to EN catalytic function. Taken together, our results revealed the structural characteristics of EBIV EN and made important implications for understanding the catalytic mechanism of cap-snatching ENs.

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Address correspondence to Peng Gong, gongpeng@wh.iov.cn, or Yiwu Zhou, zhouyiwu@hust.edu.cn.

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The *Bunyavirales* is comprised of a diverse group of segmented negative-strand RNA viruses (sNSVs) that infect a variety of animals and plants. The majority of bunyaviruses are carried and transmitted by arthropods except for hantavirus and arenavirus (1). A number of bunyaviruses could cause severe human diseases, including the well-known Crimean-Congo hemorrhagic fever virus (CCHFV), Rift Valley fever virus (RVFV), and Lassa virus (LASV). Currently, no vaccines or effective drugs against these viruses are available. A typical genome of bunyavirus comprises three single-stranded RNA segments, namely, large (L), medium (M), and small (S) based on their relative size. The L segment encodes the L protein, also known as the RNA-dependent RNA polymerase (RdRP). The M segment encodes the precursor of viral glycoprotein that is further cleaved into the mature glycoproteins Gn and Gc by cellular proteases. The S segment encodes the nucleocapsid (N) protein. Each genomic segment inside the bunyavirus virion is coated with multiple N proteins and associated with the RdRP to form a ribonucleoprotein complex (RNP), which is the basic functional unit for viral transcription and replication (2).

Besides the polymerase core, some viral RdRPs also contain functional domains related to the transcription initiation. The sNSVs, including bunyaviruses and orthomyxoviruses, employ a unique "cap-snatching" mechanism to initiate viral transcription. As demonstrated in the influenza viruses of *Orthomyxoviridae*, the RdRP complex first binds to the host-derived capped mRNA via its cap-binding subunit PB2; then, the capped mRNA is cleaved at 10 to 15 nucleotides (nts) downstream of the cap site by the endonuclease (EN) subunit PA, and the resulting short capped RNA fragments are used as primers to initiate viral mRNA transcription by the RdRP catalytic subunit PB1 (3, 4). The cap-snatching EN module universally exists in the RdRP of sNSVs and likely shares common features in substrate recognition, catalysis, and coordination with RdRP core and the cap-binding module (5). In bunyaviruses L protein, the RdRP core is flanked by the N-terminal EN and C-terminal cap-binding modules (6).

The EN domain of bunyaviruses and orthomyxoviruses, which comprises approximately 200 residues, is a metal ion-dependent nuclease and harbors a highly conserved PD. . .D/ExK catalytic motif (5). Although the sequence homology was low, the available crystal structures of EN from different sNSVs showed a similar overall architecture consisting of central antiparallel β-sheets surrounded by multiple α-helices (7–13). The conserved active site is located at the center of the EN domain and is capable of divalent metal-ion binding. However, some divergences occur in the active sites of available EN structures; first, the numbers (from zero to two) of bound cations vary in different EN apo structures, and in some studies, there are even contradictory observations (9, 12). Additionally, the binding mode of the metal ions with the active site residues differs and is closely related to the level of EN activity (9). In general, the structures complexed with the substrate may reflect the actual metal-ion binding state in the EN catalytic state. However, in two recently reported structures of EN from influenza A virus (IAV) in complex with DNA oligomer, the number of bound cations are also different (14). Owing to the lack of structural information on EN-substrate complexes in a catalytically competent state, the exact mechanism and number of metal ions required for catalysis remain elusive. The cap-snatching EN provides an ideal target for developing broad-spectrum antivirals against sNSVs owing to conserved catalytic motifs as well as the overall fold. Currently several inhibitors against IAV EN have been reported, among which the baloxavir marboxil (BXM) is the first U.S. Food and Drug Administration (FDA)-approved drug treatment for influenza (15), and it also shows potent inhibitory effect against orthobunyaviruses cap-snatching EN (16). Furthermore, the diketo acid compounds, especially the L-742,001, have been demonstrated as a potent EN inhibitor against both influenza viruses and various bunyaviruses (17, 18).

The Ebinur Lake virus (EBIV) is a recently identified mosquito-borne bunyavirus belonging to *Peribunyaviridae*. Sero-epidemiological survey showed that local residents

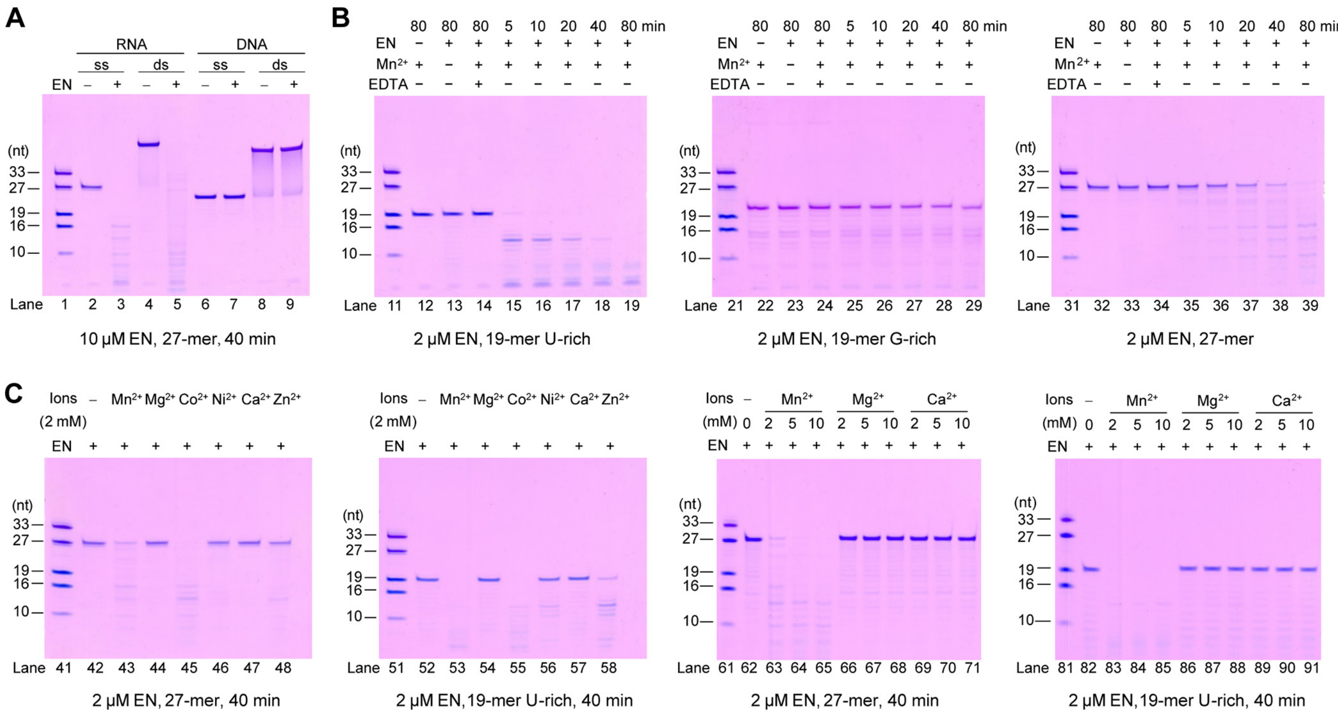

FIG 1 Enzymatic activity of the EBIV EN. (A) Nucleic acid substrate preference of the EBIV EN. Four nucleic acid substrates of identical reference sequence and length but in different forms (single- or double-stranded; RNA or DNA) were incubated with or without the enzyme in the presence of Mn^{2+}. (B) RNA sequence preference of the EBIV EN. The enzyme was incubated with 19-mer U-rich, 19-mer G-rich, or 27-mer heterogeneous ssRNAs in the presence of Mn^{2+}. For comparison, the indicated ssRNAs were incubated with Mn^{2+} only (lane 12); EN only (lane 13); or EDTA, EN, and Mn^{2+} (lane 14). (C) Divalent cation dependency of the EBIV EN. The enzyme was incubated with the 27-mer or 19-mer ssRNAs in the presence of various divalent cations. An incubation lacking divalent cations was performed for comparison.

carried antibodies against this virus, and EBIV can cause cytopathic effects in a variety of vertebrate cell types and is highly lethal to wild-type mice (19, 20). These studies suggest that EBIV is a neglected, potential pathogen in animals and/or humans. In this study, we reported a high-resolution crystal structure of EBIV EN wild-type (WT) and characterized its structural features and EN activity. To better understand the metal-ion-dependent catalysis of ENs, a series of active site point mutants were generated, and their EN activity and crystal structures were studied. Our results showed that EBIV EN is a specific RNA nuclease with substrate preference. The catalytic activity of EBIV EN is highly dependent on both metal-binding sites within the active site, and the binding of manganese ions at these two binding sites are correlated. Our work provides a basis to the understanding of sNSV EN catalysis and will benefit the development of EN-targeting compounds that have broad-spectrum potential.

RESULTS

Characteristics of EBIV EN activity. To characterize the activity of EBIV EN, we conducted a series of *in vitro* tests following the methods described in previous studies (7, 8, 10). Initially, the substrate selectivity of EBIV EN was determined using four kinds of substrates, ssRNA (structured), dsRNA, ssDNA, and dsDNA, with the same 27-mer reference sequence (10). Manganese ions (Mn^{2+}) were chosen based on the divalent cation-dependent activity of EN (8). It turned out that EBIV EN selectively cleaved RNA (single strand [ss] or double strand [ds]) but not DNA (ss or ds), indicating that EBIV EN is indeed an RNA-specific EN (Fig. 1A). Next, the RNA substrate preference was characterized by comparing the EN activity on the 19-mer uridine (U)-rich, 19-mer guanidine (G)-rich, and 27-mer heterogeneous ssRNA substrates, while the U or G nucleotides were designed at identical positions for U-rich and G-rich constructs. The results showed that the U-rich ssRNA was almost completely digested into small fragments by EBIV EN within 5 min (Fig. 1B, lane 15), whereas the majority of the G-rich ssRNA remained intact until 80 min (Fig. 1B, lane 29). Similarly, complete digestion of the structured 27-mer ssRNA only occurred after 80 min (Fig. 1B, lane 39). Together, these data suggest that EBIV EN has a substrate preference for U-rich and unstructured ssRNA. Furthermore, the specificity of EBIV EN for divalent cations was analyzed using 27-mer and 19-mer U-rich ssRNA as substrates. EBIV EN exhibited clear nuclease activity with Mn^{2+} and Co^{2+} when using both RNA constructs (Fig. 1C, lanes 43, 45, 53, and 55). However, weak and moderate activity was observed with Ni^{2+} and Zn^{2+}, respectively, only when the more sensitive substrate 19-mer U-rich ssRNA was used (Fig. 1C, lanes 56 and 58). No apparent EN activity was detected with Mg^{2+} or Ca^{2+} for both RNA constructs (Fig. 1C, lanes 44, 47, 54, and 57), albeit the concentration of metal ion was increased from 2 mM to 10 mM (Fig. 1C, lanes 66 to 71, and 86 to 91).

EN structures of sNSVs are conserved in domain architecture but are diverse in the conformation of the LA/LB loop pair adjacent to the active site. To structurally characterize the EBIV EN, we determined a WT EN crystal structure at 2.11-Å resolution, with four EN molecules in the crystallographic asymmetric unit (Table 1). The global conformations of these four molecules are consistent with root-mean-square deviation (RMSD) values of 0.4 to 0.6 Å for all superimposable α-carbon atoms (chain B as reference, 98% to 99% residue coverage). The structure of EBIV EN also exhibited similarities with other cap-snatching ENs from several families of *Bunyavirales* and *Orthomyxoviridae* (RMSD 1.2 to 3.2 Å, chain B of EBIV EN as reference, 58% to 97% residue coverage), with the highest similarity to the LACV EN from the same virus family (Fig. 2A). Structural comparison revealed that the ENs of various sNSVs had a conserved architecture composed of two lobes. One lobe consists of a central β-sheet surrounded by a long α-helix and several short helices, and the other lobe is mainly composed of a helix bundle. The EN active site is located between these two lobes and normally harbors one or two catalytic divalent cations coordinated by highly conserved residues. Surface electrostatic potential analysis showed that the active site of EBIV EN was highly negatively charged, which may contribute to the binding of catalytic divalent cations, whereas some positively charged regions around the active site may accommodate the binding of RNA substrates (Fig. 2B).

Despite the conservation of domain architecture, some structurally diverse regions were noted in the structures of different viral ENs, in particular the two loops adjacent to the active site: loops A (LA) and B (LB) (Fig. 2A). The LA loops from different EN structures were similar in length but distinct in sequence and conformation. Two conformational states for the LA loop could be distinguished from different EN structures: (i) the "closed" state, with the LA loop in proximity to the active site and (ii) the "open" state, with the LA loop away from the active site. A relatively conserved acidic residue on the LA loop is present in all reported structures, equivalent to N52 of EBIV EN, which is involved in the coordination of Mn1 (denoted as metal site 1). In EBIV, Hantavirus (HTNV), and IAV, the LA loops adopt a closed conformation, enabling the N52 in EBIV and the functionally equivalent glutamate (E) in the other two viruses to approach the active site and coordinate with Mn1. By contrast, in LACV, severe fever with thrombocytopenia virus (SFTSV), and LASV, the LA loops adopt the open conformation; thus, their corresponding aspartate residue (D) is away from the active site and unable to coordinate with Mn1. The LB loop is adjacent to the active site, with an absolutely conserved PD sequence (residues 78 to 79 in EBIV EN) at its C-terminus, and the conserved aspartate is responsible for the coordination of both Mn1 and Mn2 (denoted as metal site 2) (7, 9). Apart from the conserved residues, the length, sequence, and conformation of the LB loop are quite diverse across different EN structures. For instance, the loop is the shortest in the IAV EN structure, whereas it is the longest in the HTNV EN structure owing to an insertion at the N-terminal region (Fig. 2C and Fig. 3). Overall, the LA/LB loop pair contains highly conserved active site residues responsible for Mn^{2+} coordination at the two metal-ion-binding sites, indicating its pivotal role in EN activity. Moreover, the LA/LB loop pair adopts diverse conformations in different EN structures, implying its specific roles in regulating EN activity.

TABLE 1 X-ray diffraction data collection and structure refinement statistics

	EBIV EN construct (PDB ID code)							
Parameter	WT (7EWU)	H34A (7EWV)	N52A (7EWW)	P78A (7EWX)	D79A (7EWY)	D92A (7EWZ)	K108A (7EX0)	Y109A (7EX1)
Data collection								
Space group	$P2_1$	$P2_1$	$P2_1$	$P2_1$	P1	$P2_1$	$P2_1$	$P2_1$
Cell dimensions								
a, b, c (Å)	70.8, 41.6, 145.3	70.8, 41.0, 75.0	40.0, 70.8, 70.3	69.3, 41.4, 85.8	41.2, 69.7, 72.8	40.9, 72.2, 71.9	69.8, 41.5, 85.8	71.5, 40.8, 74.6
α, β, γ (°)	90.0, 101.6, 90.0	90.0, 99.8, 90.0	90.0, 89.5, 90.0	90.0, 104.9, 90.0	90.1, 89.9, 90.0	90.0, 90.6, 90.0	90.0, 104.7, 90.0	90.0, 100.9, 90.0
Resolution (Å)[a]	50.00 to 2.11 (2.19 to 2.11)	50.00 to 2.61 (2.70 to 2.61)	50.00 to 2.34 (2.61 to 2.34)	50.00 to 1.95 (2.02 to 1.95)	69.64 to 2.01 (2.06 to 2.01)	50.00 to 2.05 (2.12 to 2.05)	50.00 to 2.07 (2.14 to 2.07)	50.00 to 2.30 (2.38 to 2.30)
R_{merge}	0.085 (0.566)	0.086 (0.437)	0.088 (0.599)	0.071 (0.345)	0.101 (0.678)	0.078 (0.549)	0.040 (0.455)	0.091 (0.483)
R_{meas}	0.096 (0.636)	0.099 (0.508)	0.100 (0.681)	0.085 (0.416)	0.143 (0.959)	0.087 (0.605)	0.047 (0.533)	0.106 (0.557)
$CC_{1/2}$	1.012 (0.848)	0.991 (0.795)	0.978 (0.751)	0.985 (0.820)	0.990 (0.545)	1.000 (0.910)	1.000 (0.837)	0.997 (0.895)
$I/\sigma I$	13.4 (2.7)	10.1 (2.0)	11.8 (2.2)	12.7 (2.5)	6.9 (2.1)	18.6 (2.5)	29.2 (2.8)	11.5 (2.3)
Completeness (%)	97.9 (99.2)	88.2 (89.1)	94.1 (97.5)	96.3 (99.1)	98.0 (96.8)	94.5 (76.5)	99.8 (99.7)	97.0 (98.6)
Redundancy	4.4 (4.6)	3.4 (3.3)	4.2 (4.3)	3.1 (3.0)	3.6 (3.1)	5.2 (4.7)	3.6 (3.6)	3.9 (3.8)
Refinement								
Resolution (Å)	2.11	2.61	2.34	1.95	2.01	2.05	2.07	2.30
No. reflections	47,096	11,662	15,589	33,458	52,315	24,847	29,287	18,543
R_{work}/R_{free} (%)	24.0/28.5	22.6/24.9	20.8/24.6	20.1/24.3	20.5/24.3	22.7/27.1	20.3/23.0	22.0/25.6
No. atoms								
Protein	5,687	2,652	2,660	2,835	5,368	2,717	2,836	2,655
Ligand/ion/water	−/6/176	−/2/25	−/2/51	−/5/216	−/−/284	−/4/52	−/4/191	−/2/59
B-factors (Å^2)								
Protein	45.3	48.8	57.5	35.6	33.7	52.0	41.6	48.3
Ligand/ion/water	−/63.5/43.2	−/103.2/42.1	−/67.1/56.5	−/39.7/39.9	−/−/38.4	−/65.9/53.3	−/79.5/43.9	−/32.0/48.8
RMSD								
Bond lengths (Å)	0.008	0.010	0.008	0.008	0.007	0.009	0.007	0.009
Bond angles (°)	0.960	1.035	0.983	0.895	0.852	0.997	0.876	0.940
Ramachandran statistics[b]	90.1/9.4/0.5/0.0	88.2/10.9/1.0/0.0	90.4/8.9/0.0/0.6	92.6/7.1/0.3/0.0	93.2/6.6/0.2/0.0	91.4/8.3/0.3/0.0	91.9/7.5/0.6/0.0	91.2/8.5/0.3/0.0

[a]Values in parentheses are for highest-resolution shell.
[b]These values are expressed in percentages for "most favored/additionally allowed/generously allowed/disallowed" regions in Ramachandran plots.

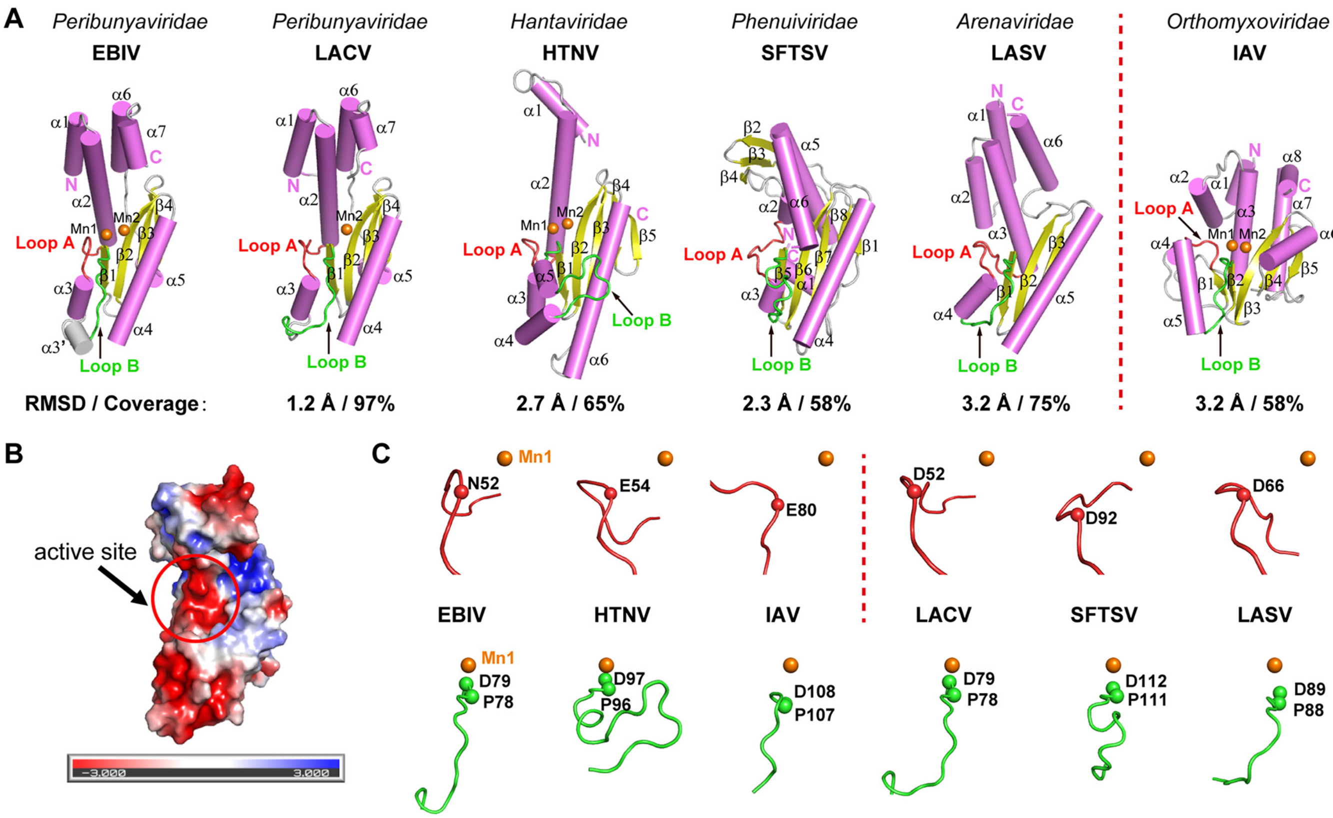

FIG 2 Crystal structure of EBIV EN and comparison with other sNSV cap-snatching ENs. (A) Comparison of the overall structure of ENs from *Bunyavirales* and *Orthomyxoviridae*. The global structures of ENs are represented in cartoon with helices shown as cylinders. The secondary structure elements were labeled and colored as follows: α-helices in violet, β-sheets in yellow, and loops in gray. The LA and LB loops are highlighted and colored in red and green, respectively. Bound cations in the active site are shown as orange spheres. The RMSD and related residue coverage values are indicated using chain B of EBIV EN as reference. PDB entries: 7EWU (EBIV); 2XI5 (LACV); 5IZE (HTNV); 6NTV (SFTSV); 5IZH (LASV); 2W69 (IAV). (B) Electrostatic surface potential of the EBIV EN. The active site with a high negative potential (red) is marked by a red circle. Scale bar: -3 kT/e in red to $+3$ kT/e in blue. (C) Structural comparison of the LA/LB loop pair of sNSV ENs. In the top row, the LA loops of EBIV, HTNV, and IAV ENs adopt a closed conformation (left); those of LACV, SFTSV, and LASV ENs adopt an open conformation (right). In the bottom row, the LB loops from different sNSV ENs exhibit diverse conformations. The Mn1 of EBIV EN (colored in orange) is shown in all structures as a control. The α-carbon atoms of conserved active site residues on LA/LB loop pairs are shown as spheres.

Structural differences in the active site of EBIV and LACV EN. Both EBIV and LACV belong to the *Orthobunyavirus* genus of *Peribunyaviridae* family, and the overall structure of EN from these two viruses are quite similar; thus, we further compared the structural details of the active sites of these two ENs. EBIV EN bears a conserved active site motif H. . .N. . .PD. . .DxK. . .T. . .KY, which is also presented in LACV EN (except the N is replaced by a D) (7, 8) (Fig. 3). Nevertheless, a significant discrepancy between the two apo structures was found in the number of Mn^{2+} bound in the active site and the conformation of the LA loop. In the apo EBIV EN structure, the LA loop had a closed conformation, and two manganese ions were observed in the active site. Mn1 was octahedrally coordinated by the side chains of N52 of the LA loop, D79 of the LB loop, and four water molecules (Fig. 4A), whereas Mn2 was only coordinated by the main chain carbonyl of Y93 of β2, the side chain of D92, and one water molecule. The electron densities and number of ligands for the two manganese ions suggest strong and weak binding for Mn1 and Mn2, respectively, in this structure. Conversely, in the LACV EN apo structure, the LA loop has an open conformation; Mn1 is absent in the active site, whereas Mn2 is bound to six ligands containing the side chains of D79 of the LB loop, H34 on α2, D92 on β2, and the main chain carbonyl of Y93 of β2, together with two water molecules (Fig. 4A). Nevertheless, the biochemical data in the LACV EN study suggest that Mn1 can still bind to the active site (8). Because the structurally observed

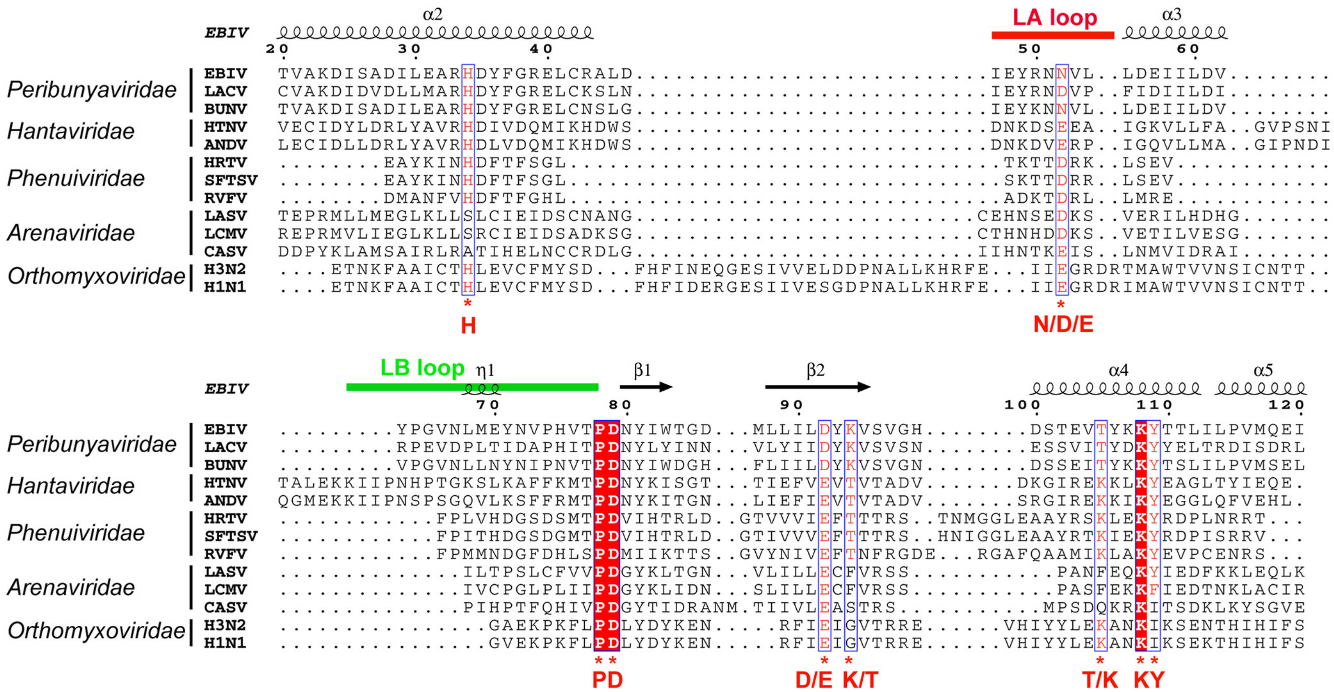

FIG 3 Structure-assisted sequence alignment of the ENs from *Bunyavirales* and *Orthomyxoviridae*. The EN protein sequences were aligned with the program ClustalW and then manually adjusted. The key conserved active site residues are highlighted with red letters and labeled with asterisks. LA and LB loops are indicated with red and green straight lines, respectively. The secondary structure of EBIV EN is shown over the sequence alignment with helices and β-strands shown as springs and arrows, respectively. The identical residues are colored in white on a red background, the relatively conserved residues are colored in red and framed in blue. BUNV, Bunyamwera virus; ANDV, Andes virus; HRTV, Heartland virus; LCMV, Lymphocytic choriomeningitis virus; CASV, California Academy of Sciences virus; H3N2, influenza A virus, subtype H3N2; H1N1, influenza A virus, subtype H1N1.

binding of the two manganese ions is inconsistent between these two highly similar apo EN structures, we hypothesize that the binding and coordination of the two divalent cations in the EN active site might be a dynamic process depending on the active site conformation and RNA substrate binding.

Mutations of the conserved residues in active site impair EBIV EN activity. To analyze the functions of the active site residues in the catalytic activity of EBIV EN, we performed single-point alanine mutations at nine conserved residues: H34, N52, P78, D79, D92, K94, T105, K108, and Y109 (Fig. 3). Among these, six residues (H34, N52, P78, D79, D92, and Y109) are involved in Mn1 or Mn2 binding. The K108 and the two relatively conserved residues (K94 and T105) are not associated with metal ion coordination (Fig. 4A). The D24 of α2 was chosen as a negative control because this residue is distant from the active site but is also conserved in *Peribunyaviridae*. All these mutants were successfully expressed and prepared with high purity (Fig. 4B). As expected, a single mutation in the active site residues involved in Mn1 (N52, P78, D79, D92, and Y109) and Mn2 (H34, D79, and D92) coordination abolished EN activity (Fig. 4C). This result supports the fact that the above residues engage in Mn1 and Mn2 coordination either directly or indirectly (Fig. 4A). Mutation of K94 also impaired EN activity, supporting its role as a putative catalytic residue, as described in previous studies (8). The K108A mutant exhibited low EN activity only at high protein concentrations, indicating that it is important but not essential for EN activity. The mutations at T105 and D24 (negative control) had no impact on EN activity owing to their absence in metal ion coordination. Taken together, the conserved residues in the active site coordinated with Mn1 and Mn2 are essential for EN catalytic activity, suggesting that both manganese ion binding sites contribute to EBIV EN activity.

Structural analyses of mutants support the two-metal-ion catalytic mechanism of EN and suggest the correlation of metal binding at the two binding sites. To understand how the residues of the active site affect EBIV EN activity on a structural basis, we attempted to crystallize all nine mutants and eventually solved the structure for

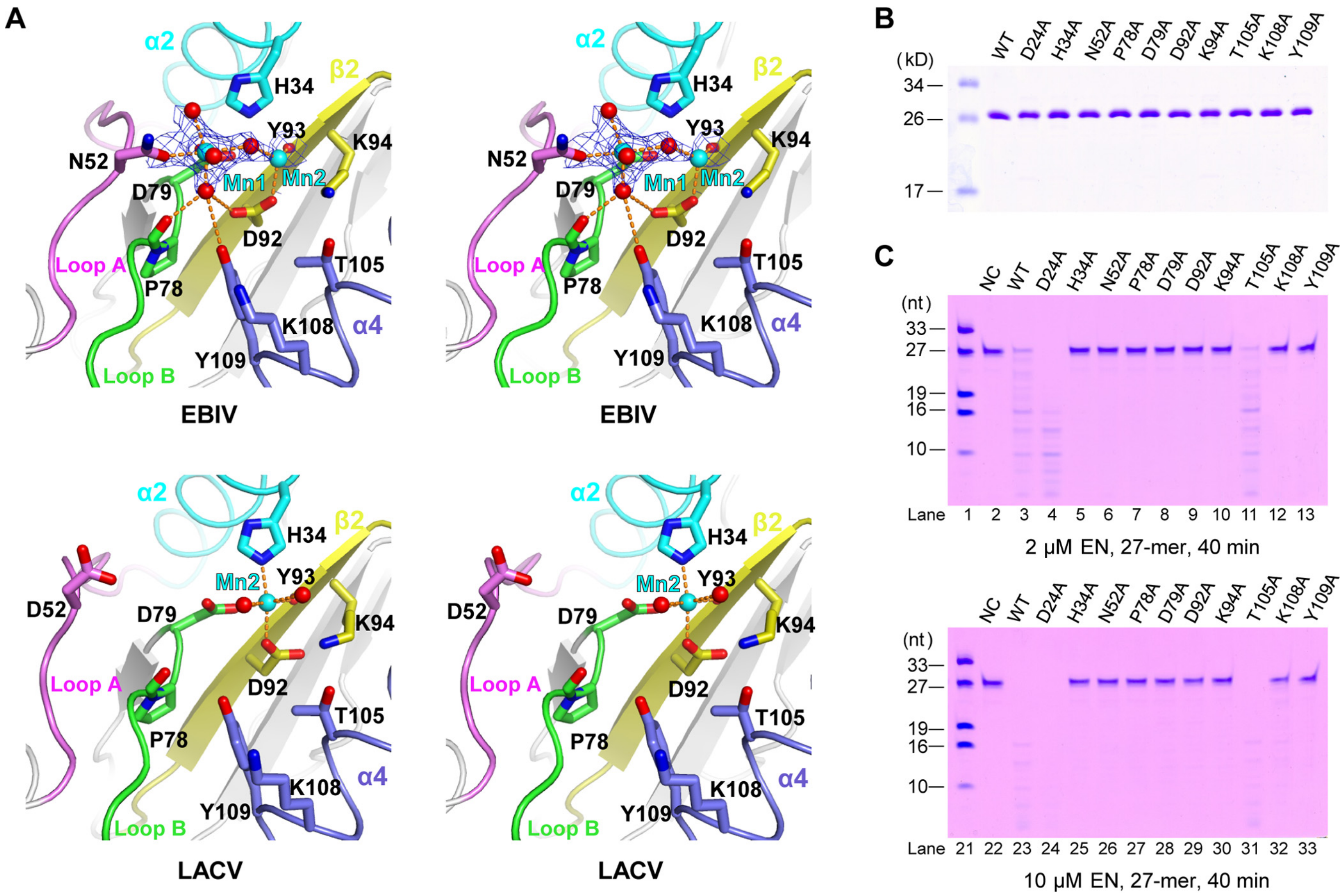

FIG 4 Comparison of structural details of EBIV and LACV EN active site, and *in vitro* EN activity analysis of EBIV WT and active site mutants. (A) Stereo-pair images of EBIV and LACV ENs active site comparison. Protein structures are shown using ribbon style with only the β-sheets shown in cartoon style. Key conserved active site residues are labeled and shown as sticks. The manganese ions (Mn^{2+}) and coordinated water molecules are shown as cyan and red spheres, respectively. Coordination interactions of manganese ions are shown as orange dotted lines. Coloring scheme: loop A in violet, loop B in green, α2 helix in cyan, α4 helix in slate, and β2 sheet in yellow. In EBIV EN, the 3,500 K composite simulated annealing (SA) omit $2F_o$-F_c electron density map (contoured at 0.7 σ) is overlaid for manganese ions and neighboring atoms participating in coordination. (B) Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) analysis of the purified WT EBIV EN and mutants. The purified proteins were loaded at an equal molar amount, separated by SDS-PAGE and stained with Coomassie brilliant blue R-250. (C) EN activity of EBIV WT and active site mutants. The 27-mer ssRNA was incubated with 2 or 10 μM WT or indicated mutant protein in the presence of Mn^{2+}. As a negative control (NC), the ssRNA was incubated with the WT protein in the absence of Mn^{2+}.

seven of them (Table 1). Among the residues mutated, H34, N52, D79, and D92 are the direct-interacting ligands of Mn1 or Mn2 coordination, P78 and Y109 are the indirect-interacting ligands of Mn1 coordination mediated by a bridging-water molecule, whereas K108 is irrelevant to metal coordination (Fig. 4A and Table 2). Comparative

TABLE 2 Manganese ion coordination analysis of EBIV EN structures

Constructs	Mn^{2+} coordination involvement in WT[a]	Bound Mn^{2+} (coordination no.)[b]	Activity[c]
WT	NA[d]	Mn1 (6), Mn2 (3)	+++
H34A	Mn2	Mn1 (2)	−
N52A	Mn1	Mn2 (3)	−
P78A	<u>Mn1</u>	Mn2 (6)	−
D79A	Mn1, Mn2	No ions	−
D92A	<u>Mn1</u>, Mn2	Mn1 (2)	−
K108A	Not involved	Mn1 (6), Mn2 (2)	+
Y109A	<u>Mn1</u>	No ions	−

[a]The underline indicates indirect involvement in manganese ion coordination.
[b]The coordinated numbers of manganese ions are counted based on the coordination bonds with distance within 2.5 Å and the electron density of corresponding ligands in the structures.
[c]The EN activity of WT and mutants is represented as "+++," "+" and "−," denoting apparent, weak, and inconspicuous activities, respectively.
[d]NA, not applicable.

structural analysis indicated that the mutations in the active site have no impact on the overall structure of EBIV EN, with RMSD values ranging from 0.7 to 1.1 Å (chain B of WT EN as reference, 93% to 99% residue coverage). However, the number of bound manganese ions (supported in part by anomalous signal) and the conformations of residues at the active site of the mutants differed, if compared with that of the WT structure. Among these seven mutant structures, four bound only one metal ion (H34A, N52A, P78A, and D92A), and two bound none (D79A and Y109A). Only the K108A mutant which is not involved in metal binding still bound two metal ions (Fig. 5A and Table 2). These structural data support the results of the EN activity assay (Fig. 4C) and the two-metal-ion catalytic mechanism of EBIV EN. Considering that both Mn^{2+} and Mg^{2+} were present in the crystallization drop, we cannot rule out the contribution of Mg^{2+} to the $2F_o$-F_c electron density map of the structures. Thus, the mixed binding in the active site of WT or mutant structures is possible.

Except for the changes in the number of bound manganese ions, the binding state of manganese ions at the two metal sites was clearly altered in some mutant structures. Of note, the mutation of P78 associated with Mn1 coordination resulted in apparently stronger binding of Mn2 (fully coordinated by six atoms) (Fig. 5B) than that (coordinated by three atoms) in the apo structure (Fig. 4A). Meanwhile, changes in active site conformation occurred in the P78A mutant. For example, the main chain carbonyl of the neighboring T77 rotated and the α-carbon atom of residue N52 of the LA loop underwent about 1 Å movement away from the active site, resulting impairment of the Mn1 coordination. Besides, the side chain of D92 rotated and participated in Mn2 coordination. Similarly, the mutation of Y109 associated with Mn1-coordination affects the binding state of Mn2, but the result is an elimination of its coordination owing to a rotation of the side chain carboxyl group for D92 in the structure (Fig. 5A). Taken together, these structural data indicate that the binding of manganese ions at the two metal binding sites is likely correlated and regulated by the conformations of the LA/LB loop pair along with the active site residues.

DISCUSSION

The EN encoded by sNSVs is essential for initiating viral transcription due to its key role in the cap-snatching process. In this study, we biochemically characterized the EBIV EN activity using different kinds of nucleic acid substrates. It showed that EBIV EN preferably cleaves RNA rather than DNA (Fig. 1A), this may be related to the specificity and efficiency of the substrate binding. A recent work on the IAV EN showed that the 2′-OH of the RNA ribose group could provide additional interaction with its binding site to optimize the binding and cleavage efficiency (14), suggesting RNA is the natural substrate for EN. Our data also showed that EBIV EN favors unstructured U-rich ssRNA (Fig. 1B), albeit this is inconsistent with a previous study in which the IAV, LACV, and HTNV ENs are able to cleave both U and G- rich RNA efficiently (9). Several divalent metal ions including Mn^{2+}, Co^{2+}, Zn^{2+}, and Ni^{2+} are able to activate the EBIV EN, no apparent activity was observed in the presence of Mg^{2+} and Ca^{2+} (Fig. 1C), although both of them are among the most abundant divalent metal ions in cells. These results are largely consistent with previous studies, having most of the ENs require Mn^{2+} for *in vitro* activity (7–9), and IAV and LASV ENs show low levels of activity in the presence of Mg^{2+} (12, 21). The activity of EN might be associated with the properties of divalent metal ions, such as atomic (covalent) radius and coordination characteristics, as well as the affinity of EN active site to metal ions. For example, the Mn^{2+} showed more "relaxing" substrate specificity than Mg^{2+} perhaps due to its less stringent coordination requirements (22). Even now the question about which divalent metal ion is employed by viral ENs in cells is still unclear and needs further studies.

Many nucleases that cleave DNA or RNA require divalent metal ion cofactors for activity and exhibit diversity in their structures, catalytic motifs, and metal-ion dependent mechanisms. The majority of nucleases can be divided into three major categories based on their metal-ion dependence: one-metal-ion, two-metal-ion, and metal-independent catalysis (23). Nucleases belonging to the same structural superfamily may use different

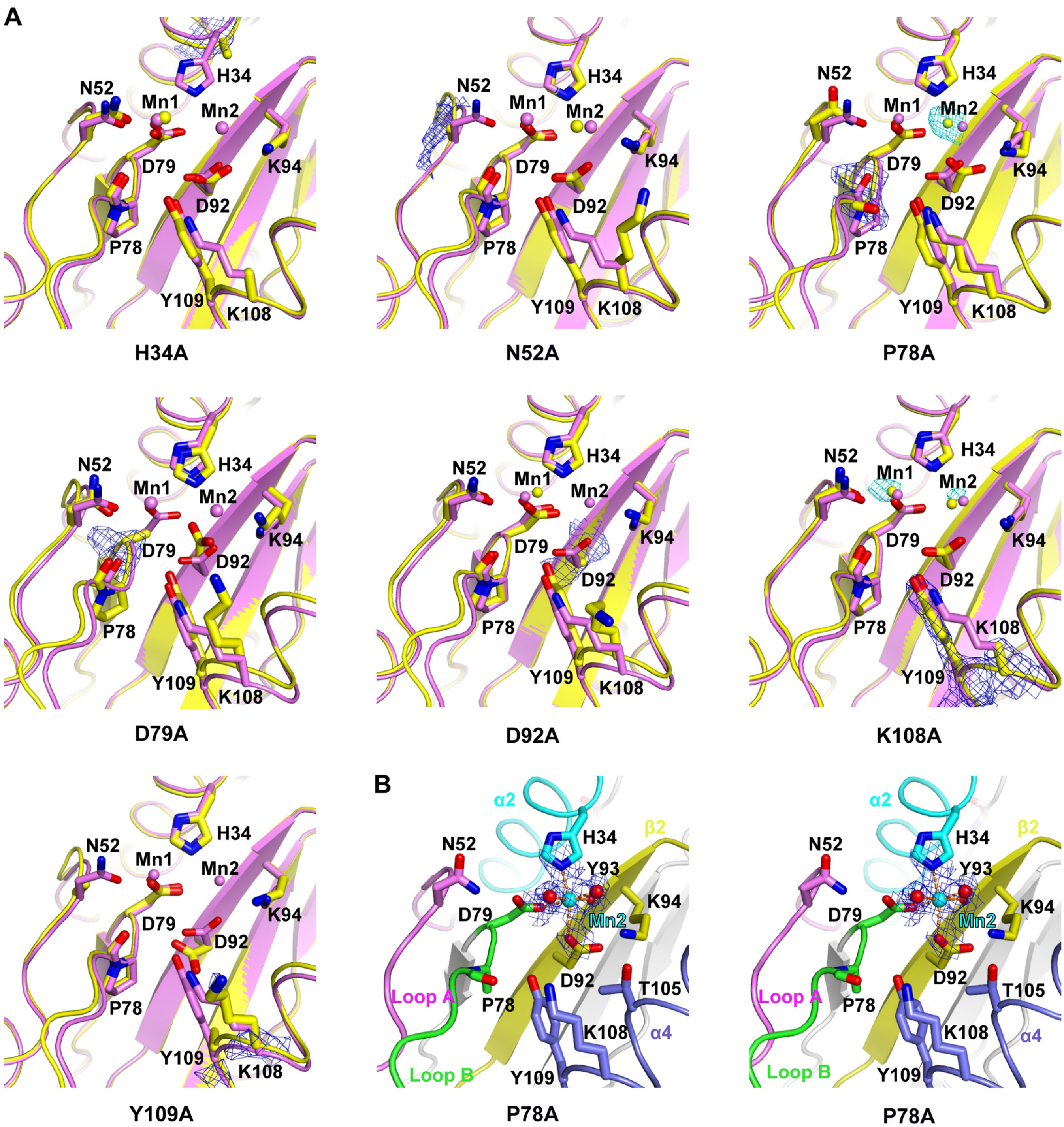

FIG 5 Crystallographic characterization of EBIV EN mutants. (A) Superimposition of WT EBIV EN and mutant structure pairs zoomed into the active site. WT EBIV EN and mutants are represented using ribbon/cartoon style and colored in violet and yellow, respectively. Key conserved active site residues are labeled and shown as sticks. A 3,500 K composite SA-omit electron density map (contoured at 1.5 σ and colored in blue) is shown for the mutated residues in mutant structures. The anomalous difference map (contoured at 3 σ and colored in cyan) is overlaid for manganese ions in some mutant structures and calculated with PHENIX software using data processed by HKL2000 in anomalous mode. (B) Stereo-pair images of P78A mutant active site with the 3,500 K composite SA omit $2F_o$-F_c electron density map (contoured at 0.8 σ) overlaid for Mn-2 and neighboring atoms participating in coordination. Coloring scheme is the same as in Fig. 4A.

catalytic mechanisms. For example, one- or two-metal-dependent mechanisms are found among the type II restriction endonucleases with a conserved PD-(D/E)xK catalytic motif (24, 25). The cap-snatching endonuclease of sNSVs was firstly found in IAV and identified as a member of the PD-(D/E)xK nuclease superfamily. The sNSV ENs were classified into two groups according to the presence or absence of a conserved histidine upstream of the catalytic motif, named His+ENs and His−ENs, respectively (9). The His+ENs show high *in vitro* activity in the presence of divalent cations, and a two-metal-ion catalytic mechanism was proposed based on the crystal structure with divalent cations and on functional studies characterizing the active site mutants (7, 8). In contrast, the His−ENs are inactive or show low *in vitro* activity and its activation mechanism remains unknown. Recently, two IAV EN (belonging to His+ENs) structures complexed with DNA oligomer were reported, however, one of them binds two Mn^{2+} ions in the active site and the other one bind a single Mn^{2+} ion (14). It still remains to be clarified whether both of the two metal ions participate in catalysis. Our study showed that the EBIV EN exhibits high *in vitro* activity and belongs to His+ENs. A single mutation at the active site residues involving in both Mn^{2+} ions coordination, resulted in nearly complete abolishment of EBIV EN activity (Fig. 4), suggesting two metal ions are required for EN activity. Structural characterization of these inactive mutants further found some of them bound with only one or no metal ions, whereas none bound with two metal ions simultaneously (Fig. 5A and Table 2). These structural data provide key insights to better interpret the enzymatic data from mutant endonucleases, supporting the two-metal-ion catalytic mechanism of EBIV EN.

Previous studies showed that at least two metal binding sites were observed in different cap-snatching EN crystal structures, whereas the binding states of metal ions at the two sites are diversified. For EN apo structures, some of them bind metal at Mn1-site (26), some bind metal at Mn2-site (8, 10, 11), and others bind metal ions at both sites (7, 9). The relative binding strengths of the metal ions at the two sites varied in different structures. Nonetheless, metal ions bound at both sites were universally observed in the EN structures upon inhibitor binding (8, 11, 27–29), and in most cases, the closed conformation of the LA loop could be induced by the inhibitors and contributed to the Mn1 coordination. In this study, we showed the binding state of manganese ions at the two sites in EBIV EN is different from that of LACV EN (Fig. 4A). Moreover, we found that the perturbation in metal coordination at one binding site could modulate the binding of metal ions at the other site. For example, the mutation of residues involved in Mn1 coordination resulted in enhancement (P78A) or elimination (Y109A) of metal-ion binding at the Mn2-site. In addition, the binding states of metal ions at the two sites were regulated by the conformational changes of the LA/LB loop pair along with the active site residues (Fig. 5 and Table 2). Collectively, these structural data strongly suggest the binding of metal ions at the two binding sites in the EBIV EN active site is correlated and dynamic, which may contribute to the functions of enzymes under different situations, such as substrate or inhibitor binding.

In conclusion, we structurally and biochemically identified the EBIV EN as a specific RNA nuclease with substrate and divalent metal ion preference, and collected evidence to support the two-metal-ion catalytic mechanism. Importantly, we found the binding of metal ions at the two sites is dynamic and correlated, implying the sensitivity and flexibility of the active site that may adapt EN to accomplish substrate binding and catalysis. Nevertheless, validation of the two-metal-ion catalytic mechanism of EN may require the determination of EN-RNA substrate complex structures at different catalytic states. The mechanistic details in EBIV EN metal binding and catalysis presented here, in combination with the high-resolution and tunable EBIV EN crystallographic system, will serve as an important basis for EN-targeting antiviral compounds against sNSVs with broad-spectrum potential.

MATERIALS AND METHODS

Protein expression and purification of EBIV EN and its mutants. The cDNA fragment encoding residues 1–183 of EBIV L protein was cloned into pET-28a(+) expression vector and fused with an N-terminal hexa-histidine tag, resulting in the plasmid pET28a-EBIV-EN. Nine EBIV EN constructs with point mutation at the conserved residues of active site (H34, N52, P78, D79, D92, K94, T105, K108, and Y109) were generated using overlap extension PCR method and the pET28a-EBIV-EN as the parental plasmid (30). A point mutation at the conserved D24 residue far away the active site was made as the negative control. All constructs were transformed into *E. coli* strain BL21(DE3) for protein expression after the verification of DNA sequencing. Cells were grown at 37°C in LB medium containing 40 μg/mL kanamycin and induced with the IPTG (isopropyl-β-d-thiogalactopyranoside) at a final concentration of 0.4 mM when the optical density at 600 nm reached at 0.6 to 0.8. The culture was grown for additional 18 h at 16°C before harvesting.

For protein purification, cells were pelleted and then resuspended in lysis buffer (50 mM Tris-Cl [pH 8.0], 300 mM NaCl, 5 mM imidazole) and disrupted by a JN-02C high-pressure homogenizer (Guangzhou Juneng Nano & Bio Technology Co., Ltd). The clarified lysate was loaded onto a self packed Sepharose column chelated with nickel ion, washed with 8 column volumes of lysis buffer containing 20 mM imidazole and then the target proteins were eluted by lysis buffer with 300 imidazole. The eluted proteins were concentrated and further purified by size exclusion chromatography using a Superdex 200 column (GE Healthcare) equilibrated with GF buffer (5 mM Tris [pH 7.5], 300 mM NaCl, 20% [vol/vol] glycerol). The purified proteins were concentrated to 15 to 25 mg/mL, flash frozen in liquid nitrogen, and stored as aliquots at −80°C. The typical yield is about 4 to 7 mg of pure protein per liter of bacterial culture.

Crystallization, diffraction data collection, and structure determination. Crystallization of the EBIV EN and its variants were performed by sitting drop vapor diffusion at 16°C. Typically, 8 and 10 mg/mL protein supplemented with 5 mM $MnCl_2$ were mixed with the precipitant/reservoir solution at a 1:1 volume ratio in 0.8-μL drops. Twin diamond-shaped crystals of WT protein appeared in the precipitant/reservoir solution (0.2 M $MgCl_2$, 0.1 M Bis-Tris [pH 5.2], 28% [wt/vol] PEG 3350) within 1 to 2 weeks. The crystals of mutants were obtained under the same crystallization condition with similar morphology as the WT, except for those of P78A, which are large flakes. The WT and mutant crystals were flashily cooled in liquid nitrogen in a cryo-solution of precipitant solution supplemented with 15% (vol/vol) glycerol and 2.5 mM $MnCl_2$. The X-ray diffraction data were collected at BL17U1 beam line of Shanghai Synchrotron Radiation Facility (SSRF) with a wavelength of 0.979 Å and temperature of 100 K. At least 180 degrees of data was typically collected in 0.2° to 0.4° oscillation steps.

For structure determination, the diffraction data sets were indexed, integrated, and scaled by HKL2000 (31) or Xia2 (32), the initial model was obtained by the molecular replacement (MR) program PHASER (33) using coordinates from La Crosse virus (LACV) EN structure (PDB entries 2XI5) as the search model. The structure refinement was performed iteratively with manual model building in Coot (34) and automated refinement by PHENIX (35). The 3,500-K composite simulated-annealing omit $2F_o$-F_c electron density maps were generated by CNS (36). The data collection and refinement statistics for the final models are listed in Table 1. Unless otherwise indicated, all the structure superimpositions were performed by the maximum likelihood-based structure superpositioning program THESEUS (37). The structure images were prepared using PyMOL (38).

***In vitro* EN activity assays.** Different kinds of substrates were designed to test the activity of EBIV EN, including 27-mer ss or ds RNA or DNA substrates (5′-GA(U/T)GA(U/T)GC(U/T)A(U/T) CACCGCGC(U/T)CG(U/T)CG(U/T)C-3′), 19-mer U-rich ssRNA (5′-AUUUUGUUUUUAAUAUUUC-3′) and 19-mer G-rich ssRNA (5′-AGGGGCGGGGGAAGAGGGA-3′). All the oligonucleotide substrates were obtained by gene synthesis (Shanghai Sangon Biotech Co., Ltd; Integrated DNA Technologies) and dissolved in RNase-free distilled water. EN activity was measured as previously described with minor modification (7, 8). Briefly, 2 or 10 μM purified EN proteins were incubated with various RNA or DNA substrates (10 μM) at 30°C in assay buffer (5 mM Tris-HCl [pH 7.5], 250 mM NaCl, 5% [vol/vol] Glycerol), in the presence or absence of 2 mM (or indicated concentration) divalent metal ions, 20 mM EDTA. The reaction was stopped by adding 2× stop loading buffer (95% formamide, 20 mM EDTA, 0.02% [wt/vol] xylene cyanol and 0.02% [wt/vol] bromophenol blue). The samples were heated at 100°C for 1 min, separated by 7 M urea, 20% polyacrylamide, Tris-borate-EDTA gel electrophoresis, and then visualized by staining with Stains-All (Sigma-Aldrich). Note that slightly different RNA degradation patterns were observed between experiments using different batches of RNA constructs with different purity levels.

Data availability. Atomic coordinates and structure factors for the reported crystal structures have been deposited with the Protein Data bank under accession numbers 7EWU (WT), 7EWV (H34A), 7EWW (N52A), 7EWX (P78A), 7EWY (D79A), 7EWZ (D92A), 7EX0 (K108A), and 7EX1 (K109A).

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